

Speech Analysis and Linguistics

Assignment-1

Deadline: 11:59pm on 21st August 2026

All questions should be answered.

1. Please use any software of your choice to address the below.

- a. Record a sentence with your own voice, identify the phonemes in the recording audio and mark the phoneme segment boundaries.
- b. Compute short-time energy and short-time ZCR of the recorded audio with 20ms window size with 10ms window shift.
- c. Compute an average of short-time energy and short-time ZCR values within each manually marked phoneme segment and compare the averaged values across the 1. vowel and non-vowel phonemes and 2. voiced and unvoiced phonemes. Comment on the observed values.

2. Consider the same sentence in Q1 and Record it in whispering and mark the boundaries of phonemes.

Highlight and discuss the below based on normal and whisper recordings

- a. What is not changed from normal to whisper? Is the message preserved in both the cases, discuss in detail?
- b. What are new information added to whisper to become normal? Is the the Voice/speaker information preserved, discuss in detail?
- c. What is the vocal chords state while whispering? What are the other states of vocal chords while speaking normally?
- d. Can all the words in English vocabulary discriminable while whispering? How about “pig” vs “big”? Why this can’t be discriminable, explain the reasons? Can you create such examples? Is your reasons consistent with observing the spectrogram of normal and whisper speech?
- e. How can a mimicry artist imitate another speaker's voice, explain clearly?

3. For each of the following sentences, identify the most relevant area of linguistic study (Pragmatics/Discourse, Phonology, Morphology, Phonetics, Syntax or Semantics) and briefly explain your choice.

- a. The word **“reusable”** consists of the prefix **“re-,”** the root **“use,”** and the suffix **“-able.”**
- b. The word **“bank”** can refer to the side of a river or a financial institution.
- c. The sentence **“The teacher explained the lesson to the students”** follows the standard subject-verb-object structure.
- d. The words **“fan”** and **“van”** differ by only one phoneme, /f/ and /v/.
- e. When someone says **“It’s getting dark in here,”** they may actually be asking someone to switch on the light.

- f. The sound [t] in "talk" is produced by placing the tongue behind the upper front teeth.
4. Compare the vowels in "beat" and "bit." What are the differences in formants do you observe?
 5. Record yourself saying the following sentence: "**Did you miss the exam?**" with four different emotions of your choice, extract pitch contour using praat software and comment on the pitch variation for each.
 6. Let $s(t) = \sin(2\pi f_1 t); 0 \leq t \leq \tau$ & $\sin(2\pi f_2 t); \tau < t \leq 2$ Else 0, where $\tau = 1\text{sec}$; $f_1 = 100\text{Hz}$; $f_2 = 300\text{Hz}$, then empirically find the τ considering amplitude spectrum of STFT with window size of 20ms.
 7. Choose one speech-processing application (which is of recent interest) such as speech recognition, speaker identification, emotion recognition, speech synthesis, voice assistants, or pronunciation assessment.
 - a. Explain the application and its purpose.
 - b. Identify and highlight the importance of the speech information/features used by the system.
 - c. Explain where phonetics, phonology, morphology, syntax, semantics, or pragmatics may contribute to the application.
 - d. Mention one limitation of the application.

Note: Submit the assignment folder in zip format and save as "rollnumber_A1.zip"

Please include your codes and plots in the submissions.

Praat: you can download from [here](#).